ICC-1 Chest: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

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2. Lung malignancies
3. COPD, asthma, wheezy chest
4. Interstitial lung disease
5. Respiratory emergencies
6. Respiratory symptoms, presentations and clinical reasoning

PART 1: LECTURE

Community-acquired pneumonia [018] (lecture and LGW)

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Definitions and pathogenesis

- Pneumonia = acute infection with clinical and/or radiological consolidation of part of one or both lungs. No consolidation on CXR = acute bronchitis (winter, viral).
- CAP = acquired in community; HAP = 48 h or more after admission; VAP = 48 h or more on ventilator. Different organisms and antibiotics.
- Routes: inhalation (viruses, Legionella, Mycoplasma); aspiration (stroke, alcohol, hypnotics; oral flora, anaerobes); colonisation (bronchiectasis, CF; Pseudomonas); blood spread (IVDU, cannula, dialysis; S. aureus, Gram-negatives).
- Commonest organisms: S. pneumoniae, atypicals, H. influenzae. Resistant-organism risk: hospital + IV antibiotics in last 90 days, previous MRSA/Pseudomonas, comorbidities.

Diagnosis and severity

- Elderly: vague symptoms (confusion, GI upset, no fever). Signs: dullness, bronchial breathing, crackles. Consolidation = bacterial; ground-glass = atypical.
- Routine: CXR (or LUS/CT), CBC, ABG, BUN (diagnosis + severity).
- IDSA/ATS severe CAP = 1 major (septic shock on vasopressors; ventilation) or 3 minor (RR ≥30, fluid-responsive hypotension, temp <36°C, confusion, WBC <4,000, platelets <100,000, BUN ≥20 mg/dL, PaO2/FiO2 ≤250, multilobar). Room air FiO2 = 0.21.
- Sputum Gram stain/culture + blood cultures only if in hospital AND severe or resistance risk. Urinary antigen: Legionella travel/outbreak or severe CAP.
- Procalcitonin: start antibiotics in confirmed CAP regardless of level.

Treatment

- CURB-65 0-1 home; 2 ward; 3-5 or severe IDSA/ATS = ICU.
- Outpatient, no comorbidity: amoxicillin, doxycycline or macrolide. With comorbidity: amox-clav/ceftriaxone + macrolide or doxycycline, or fluoroquinolone alone.
- Inpatient: beta-lactam + macrolide (or + fluoroquinolone). Add anti-MRSA (vancomycin, linezolid) or anti-Pseudomonas (pip-tazo, meropenem) only with risk factors; adjust by culture.
- Clindamycin only if abscess/empyema suspected; oseltamivir if influenza PCR positive.
- Steroid only: severe CAP + shock, P:F <300, or rapid progression (≤3 days).
- Duration: 3-4 days non-severe and stable; 5+ days severe. Stability at 72 h: T ≤37.8, HR ≤100, RR ≤24, SBP ≥90, SpO2 ≥90%, normal mentation and intake. No routine follow-up CXR if better in 5-7 days.

Special types and prevention

- Legionella: water aerosols, haemoptysis (1/3), confusion, diarrhoea, bradycardia. Klebsiella: diabetics/alcoholics, RUL, bulging fissure.
- Influenza vaccine yearly from 6 months; LAIV not in >65, pregnancy, immunosuppression (give IIV/RIV). Pneumococcal vaccine ≥65 or high-risk 2-64.

Source: Pneumonia lecture [IM-PULMO.13.1]; CAP LGW [IM-PULMO.14.2]

PART 1: LECTURE

Bacterial infections of the lower respiratory tract [020] (Moodle topic)

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Upper airway and streptococci

- Group A Streptococcus (S. pyogenes) throat infection; toxin-associated complication = scarlet fever.
- Child with fever, drooling, stridor + swollen epiglottis on lateral X-ray = acute bacterial epiglottitis; commonest cause Haemophilus influenzae.
- S. pneumoniae: alpha haemolysis on blood agar; causes CAP, meningitis, acute otitis media, septicaemia.

Tuberculosis basics

- BCG vaccine scar = protection against M. tuberculosis.
- Mycobacteria seen with Ziehl-Neelsen stain or auramine fluorescent stain.
- Rifampicin darkens (orange-red) urine.
- Isoniazid: peripheral neuropathy, hepatitis. Ethambutol: visual disturbance. Pyrazinamide: hepatotoxicity.

Hospital-acquired pneumonia

- Risks: time on ventilator, rise in gastric pH, nasogastric tube, time in hospital, extremes of age, chronic respiratory disease.
- Organisms: Staphylococcus aureus, E. coli, Pseudomonas aeruginosa, Acinetobacter.

Community-acquired pneumonia tests

- Lobar CAP: sputum culture, blood culture (aerobic + anaerobic bottles), pneumococcal urinary antigen.
- Blood culture helps for S. pneumoniae, H. influenzae, S. aureus.

Atypical pneumonias

- Mycoplasma: young adult, cough, headache, myalgia, tiredness; aerosol spread; about 15% of adult CAP.
- No cell wall: no Gram staining; resistant to beta-lactams and glycopeptides.
- X-ray variable; culture positive in only 30-60%; gold standard = fourfold rise in paired titres (e.g. 1:40 to 1:1024).
- Treat with protein/DNA synthesis drugs (tetracyclines, macrolides, quinolones); macrolides most active.
- Legionnaires' disease: travel (e.g. Spain), fever, myalgia, confusion; bilateral interstitial shadowing; sputum Gram-negative rods with many white cells. Treat macrolide ± rifampicin.

Source: Bacterial Infections of the LRT tutorial [IM-PULMO.16.7]

PART 1: LECTURE

Diagnosis of TB [058]

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Principles and radiology

- No TB without the bacillus; no healthy carriers; any material with tubercle bacilli is pathologic.
- Infection evidence: hilar nodes, calcified primary focus, positive TST. Disease needs direct or indirect evidence of active disease.
- CXR: TB mimics any chest disease; commonest = apical infiltrates/cavitation, mediastinal nodes, pleural effusion, miliary shadows.

Samples and smear

- Samples: sputum, laryngeal swab, gastric lavage, body fluids (pleural, ascitic, CSF), bronchoscopy specimens, tissue biopsy.
- Sputum: 3 consecutive early-morning samples.
- Ziehl-Neelsen: simple, rapid, cheap; misses low bacillary loads. Fluorescent: low-power large field, quicker, more positives; but costly, needs maintenance and steady power.
- False positive: food particles, precipitated stain, scratches, fibres, contamination. False negative: poor collection/storage/staining, loss of acid-fastness (sunlight, UV, heat, >1 week hot storage).

Culture and molecular tests

- Media: egg-based (Lowenstein-Jensen), agar-based (Middlebrook 7H10/7H11), liquid (7H9).
- Culture: more sensitive, species identification, drug resistance; slow (4-8 weeks), so MGIT/Bactec used (Bactec: positive if growth index ≥10).
- PCR: rifampicin resistance = rpoB; high-level isoniazid = katG; low-level isoniazid = inhA promoter.

Immune-based tests

- ADA: pleural >70 IU/L highly suggestive, <40 excludes; ratio ≥2 or lymphocyte/neutrophil >0.75 supports TB. CSF cut-off 8 U/L; ascitic >36 U/L.
- IGRA (QuantiFERON, ELISPOT): ESAT-6 and CFP-10, not shared with BCG; one visit, result in 24 h, no reader bias, no boosting; blood processed within 12 h.
- TST: 5 TU PPD (0.1 mL) intradermal, read at 48-72 h, induration in mm (0 mm if none); PPD refrigerated, dark, away from light. Shared with BCG, boosting, two visits.
- Positive TST: ≥5 mm HIV, recent contact, old healed TB on CXR, transplant/immunosuppressed; ≥10 mm recent arrivals, IV drug users, high-risk settings, lab staff, diabetes, silicosis, CRF, steroids, malignancy; ≥15 mm no risk factor.
- Latent TB: immune response without active disease; diagnose by TST or IGRA (neither clearly preferred).

Source: Diagnosis of TB lecture [IM-PULMO.17.1]

PART 1: LECTURE

Treatment of TB [094] (lecture and WHO Moodle topic)

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Aims and regimen

- Aims: cure, prevent death, relapse, transmission and drug resistance.
- First-line: isoniazid (H), rifampicin (R), pyrazinamide (Z), ethambutol (E), streptomycin (S).
- WHO new pulmonary TB: 2HRZE/4HR (6 months of rifampicin); 2HRZE/6HE phased out; daily dosing optimal. Retreatment: 2HRZES/1HRZE/5HRE.
- Intensive phase: kill dividing bacilli, non-infectious, prevent resistance. Continuation phase: kill dormant bacilli, prevent relapse.
- MDR-TB = resistant to rifampicin and isoniazid.

Monitoring and side effects

- Clinical review; sputum at 2 months and at completion; AST, ALT, bilirubin, CBC twice weekly for 2 weeks, at 1 month, then monthly.
- Major (stop and refer): skin rash (stop all); jaundice/hepatitis from H, R, Z (stop all, reintroduce one at a time); streptomycin deafness, vertigo, nystagmus, low urine output (stop S); ethambutol visual loss, red-green first (stop E).
- Minor (continue): GI upset (small meals/bedtime); pyrazinamide gout-like joint pain (NSAIDs, aspirin, paracetamol); isoniazid neuropathy (pyridoxine 50-75 mg daily) and drowsiness; rifampicin orange urine (warn); intermittent rifampicin flu-like illness (go daily).

TB with HIV

- HIV raises latent-to-active risk 12-20 times; atypical presentation, lower-zone infiltrates, false-negative smears.
- Problems: poor tolerance and adherence, MDR common; 90% of reactions in first 2 months.
- ART: start 2-8 weeks after TB treatment if CD4 <200 (WHO file: as soon as possible within 8 weeks); no need to wait for TB treatment to end. Barriers: toxicity, interactions, pill burden, IRIS.

Special groups

- Pregnancy: oral first-line drugs safe; streptomycin avoided (fetal ototoxicity). Congenital TB presents in weeks 2-3 with hepatosplenomegaly and respiratory distress.
- Renal: rifampicin safest (bile excretion); isoniazid, pyrazinamide standard dose; aminoglycosides reduced if CrCl <50; avoid cycloserine, PAS; ethambutol needs levels or 48 h interval.
- Liver: H, R, Z hepatotoxic (H/R and R/Z synergistic); monitor LFTs closely.

Source: Treatment of TB [IM-PULMO.18.1]; WHO TB treatment guidelines 4th ed. [IM-PULMO.21.7]

PART 1: LECTURE

Extrapulmonary TB [114]

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Spread and key forms

- Mostly haematogenous spread; direct extension from an adjacent organ is less common. General symptoms: fever, malaise, weight loss.
- Miliary TB: lesion erodes into a vessel. Commonest in children < 4, elderly, immunocompromised. Lungs and bone marrow most affected (anaemia, low platelets, leukemoid reaction); may present as FUO. CXR: thousands of 2-3 mm nodules evenly spread in both lungs.
- TB meningitis: most serious form; children 0-5, elderly, immunocompromised; the one form BCG is believed to prevent. Low-grade fever, unremitting headache, then drowsiness with focal signs, then coma. Focal signs = tuberculoma; stroke from vessel thrombosis.

Other sites

- Genitourinary: kidney (fever, back pain, pyuria) to bladder; men: prostate, seminal vesicles, epididymis (scrotal mass); women: salpingo-oophoritis (pelvic pain, sterility, ectopic). Psoas spread can give an anterior thigh abscess.
- Peritonitis: common in alcoholics with cirrhosis; mild or mimics acute abdomen.
- Pericarditis: from mediastinal nodes or pleura; can cause tamponade and heart failure.
- Lymphadenitis: posterior cervical and supraclavicular nodes; may form a draining fistula.
- Pott's disease: starts in a vertebral body, spreads to the next with disc-space narrowing; collapse, cord compression (paraplegia), paravertebral abscess.
- Skin: lupus vulgaris (face nodules); prosector's wart (direct inoculation). GI TB from M. bovis milk; may mimic appendicitis.

Diagnosis

- Smear, culture, biopsy/aspirate, PCR on CSF, urine, pleural, pericardial or joint fluid. Few organisms, so cultures often negative: use NAAT.
- TST may be negative at first; repeat in a few weeks. Blood culture positive in ~50% of disseminated TB (often HIV).
- TB meningitis CSF: lymphocytes, glucose < 50% of serum, high protein, ADA > 8 U/L.
- MRI preferred for spine; CT/US for abdominal and GU disease. Bone marrow or liver biopsy if miliary TB suspected and all else negative.
- Granuloma on biopsy, or positive TST/IGRA + unexplained lymphocytic fluid: treat without proof (treatment trial).

Treatment

- Same regimens as lung TB, 6-9 months; meningitis 9-12 months; bone/joint 9 months.
- In meningitis, slides replace ethambutol with streptomycin.
- Steroids in serositis (pericarditis, peritonitis, pleuritis, meningitis) to prevent fibrosis, unless resistance suspected.
- Surgery: drain empyema, tamponade or CNS abscess; close bronchopleural fistula; resect bowel; Pott's with deficit or deformity. Not needed for lymphadenitis except diagnosis.

Source: Extrapulmonary TB [114] (IM-PULMO.19.2)

PART 1: LECTURE

Non-tuberculous mycobacteria [115]

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Basics

- Mycobacteria other than M. tuberculosis and M. leprae (atypical, environmental). Spread by inhalation, ingestion or skin penetration; tap water is a major reservoir.
- Runyon groups (slow growers 1-3): 1 photochromogens (M. kansasii, M. marinum); 2 scotochromogens (M. scrofulaceum); 3 non-photochromogens (M. avium complex, MAC). Group 4 fast growers, culture within 7 days (M. fortuitum, M. chelonae, M. abscessus).
- Risk: lung disease (COPD, bronchiectasis, CF, ABPA), trauma/surgery/cosmetic procedures not responding to standard antibiotics, breast prostheses, HIV, transplant, TNF-alpha inhibitors and other immunosuppressants.

Clinical syndromes

- 85% pulmonary; rest lymph nodes, skin/soft tissue, disseminated. Indolent course; fever the commonest symptom.
- Lung disease: MAC, then M. kansasii and M. abscessus. Fibrocavitary form = upper-lobe cavities like TB; nodular bronchiectatic form = bronchiectasis + nodule clusters, often right middle lobe and lingula.
- Lymphadenitis: healthy children 1-5 years, one-sided cervical node, no systemic symptoms; MAC 80%, then M. scrofulaceum. Differentiate from TB lymphadenitis.
- Disseminated: immunocompromised; MAC 90%, the main NTM in HIV.
- Skin: nodules in a line along lymphatics; fish tank granuloma = M. marinum; Buruli ulcer = M. ulcerans.

Diagnosis

- Lung: symptoms + cavity or nodules on CXR/HRCT + other diagnoses excluded (all), plus one of: two positive sputum cultures, one positive bronchial wash/lavage, or positive lung biopsy with typical histology.
- AFB stain (fluorochrome or Ziehl-Neelsen) cannot tell MTB from NTM. Culture of tissue is optimal; PCR and biopsy help.

Treatment

- MAC lung: macrolide (azithromycin/clarithromycin) + ethambutol + rifampin, for 12 months after culture conversion.
- M. kansasii: rifampin + ethambutol + isoniazid + pyridoxine, 12 months after culture conversion.
- Surgery: drainage/debridement + > 3 months of combined drugs in some cases; resect solitary nodules; cervical lymphadenectomy cures child NTM lymphadenitis.

Source: NTM [115] (IM-PULMO.20.2)

PART 1: LECTURE

Lung abscess and bronchiectasis [118]

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Lung abscess

- Localised pus or necrotic debris in lung tissue forming a cavity with an air-fluid level.
- Routes: aspiration (commonest: dental infection, stroke, alcohol, frequent vomiting; anaerobes); inhalation/necrotizing pneumonia (S. aureus/MRSA, Strep. milleri, Klebsiella, Pseudomonas); colonisation of diseased lung (bronchiectasis, CF, bullae, infarcts); blood spread (IV drugs, cannulas, haemodialysis, abdominal or urinary sepsis).
- 90% polymicrobial, anaerobes usual (e.g. Bacteroides fragilis); TB, fungi, parasites less common.
- Acute < 6 weeks (poorly defined wall); chronic > 6 weeks (well-defined fibrous wall).
- Starts like pneumonia, then large sputum volumes; chronic: weight loss, clubbing.
- Differential of a cavity: cavitating cancer, localised empyema, cavitary pneumoconiosis, hydatid cyst.
- Tests: sputum and blood culture, CBC (anaemia, leukocytosis); bronchoscopy for foreign body or tumour, BAL, biopsy.
- Treatment: antibiotics (clindamycin, meropenem) 4-6 weeks; percutaneous US/CT-guided drainage if antibiotics fail; lobectomy if > 6 cm, symptoms > 12 weeks, failed medical treatment or suspected cancer.

Bronchiectasis: causes and features

- Irreversible abnormal dilatation of bronchi with chronic inflammation, chronic sputum, recurrent infection and airflow limitation.
- Causes: infection (pneumonia from measles or adenovirus, TB); immune (HIV, ABPA, ?lupus/RA); bronchomalacia; immotile cilia; cystic fibrosis; obstruction (cancer, nodes).
- Patterns: cylindrical, varicose, cystic. Mechanisms: atelectasis, pressure of secretions, traction fibrosis.
- Colonisers: H. influenzae (commonest), S. pneumoniae, Klebsiella, Pseudomonas (rapidly progressive, severe).
- Persistent cough, voluminous purulent sputum, haemoptysis (can be massive), clubbing, coarse crackles that change with cough.

Bronchiectasis: tests and treatment

- CXR not sensitive (ring, tramline shadows). HRCT is the gold standard (signet ring). Spirometry obstructive; sputum culture; test for cause (e.g. TB).
- Antibiotics, mucolytics, physiotherapy; ICS and bronchodilators only with asthma/COPD; lobectomy for localised disease.
- First Pseudomonas: ciprofloxacin 750 mg every 12 h for 2 weeks, then inhaled colistin for at least 3 months.
- Non-Pseudomonas with 3 or more exacerbations a year: culture-based antibiotic + long-term macrolide (azithromycin) for at least 3 months.

Source: Lung abscess and bronchiectasis [118] (IM-PULMO.15.2)

PART 1: LECTURE

Bronchogenic carcinoma [067a, 067b] (lecture and LGW)

Topic: Lung malignancies

Basics

- 95% of primary lung tumours; peak age 50-70; old smoker with chest symptoms = cancer until proven otherwise.
- Risks: smoking (pack-years; 80% of cancers in smokers), pollution, genetics, asbestos, old lung scarring.
- Central tumours: commoner, squamous cell, early mediastinal spread, cough, dyspnoea, haemoptysis, seen at bronchoscopy. Peripheral: adenocarcinoma, early pleural spread, pleurisy and chest pain.

Local spread

- Bronchial: changed cough, haemoptysis, localized wheeze, collapse, unresolved pneumonia, abscess.
- Pancoast tumour: rib erosion (shoulder pain to axilla), subclavian artery (unequal pulses), brachial plexus (hand wasting), sympathetic chain (Horner's: ptosis, miosis, anhidrosis, enophthalmos).
- Mediastinal: trachea (stridor, brassy cough), oesophagus (dysphagia), left RLN (hoarseness), phrenic (hiccups, then diaphragm palsy), SVC (facial plethora, neck veins, arm swelling, collaterals; classically SCLC).

Paraneoplastic (mostly SCLC)

- SCLC: SIADH, ectopic ACTH. Squamous: hypercalcaemia (PTH-like), hypoglycaemia. Large cell/adeno: gynaecomastia.
- Migrating thrombophlebitis, anaemia, eosinophilia; Eaton-Lambert, neuropathy, cerebellar ataxia, dermatomyositis; clubbing and hypertrophic osteoarthropathy (painful periostitis, subperiosteal new bone); acanthosis nigricans; nephrotic syndrome.

Investigations

- Bloods, calcium, glucose, sodium, ACTH; CXR, CT chest; metastasis search: abdominal US/CT, CT brain, bone scan, PET-CT.
- Tissue is the issue: bronchoscopy for central; CT/US-guided biopsy for peripheral; node biopsy; thoracoscopy if effusion.

Staging (NSCLC TNM 8th)

- T1 up to 3 cm; T2 3-5 cm (T2a 3-4, T2b 4-5) or main bronchus, visceral pleura, atelectasis; T3 5-7 cm or chest wall, phrenic, parietal pericardium; T4 over 7 cm or carina, mediastinum, heart, great vessels, oesophagus, vertebra.
- N1 ipsilateral hilar/peribronchial; N2 ipsilateral mediastinal/subcarinal; N3 contralateral, or any scalene/supraclavicular.
- M1a contralateral lung, pleural nodules, malignant effusion; M1b single extrathoracic; M1c multiple.
- Metastasis sites: adrenals, brain, liver, bone.

Treatment

- NSCLC: surgery if resectable with no metastases, plus radiotherapy/chemotherapy; overall 5-year survival ~20%.
- SCLC: limited vs extensive (not TNM); chemotherapy (very sensitive at first); surgery usually contraindicated; 10-20 months treated vs 2-4 untreated.

Source: Bronchogenic carcinoma lecture [067a] and LGW [067b]

PART 1: LECTURE

Asthma [005] (lecture, LGW and CBD)

Topic: COPD, asthma, wheezy chest

Definition and diagnosis

- Heterogeneous disease with chronic airway inflammation; defined by **variable** symptoms (dyspnea, cough, wheeze, chest tightness) plus **variable** expiratory airflow. COPD = persistent symptoms and obstruction.
- Airway hyper-responsiveness is the key feature. Typical: more than one symptom, worse at night, with viral infections, exercise, laughing, cold air, allergens.
- Asthma less likely with: isolated cough, chronic sputum, chest pain, dyspnea with dizziness/paresthesia, stridor.
- Atopy history and nasal polyps support but are not essential.

Tests

- Spirometry (may be normal outside attacks): FEV1/FVC < 80%; reversibility = FEV1 rise of **12% and 200 mL** 10-15 min after salbutamol 200-400 mcg.
- PEF diurnal variability > 10% = asthma: (highest - lowest) / mean x 100; twice daily, best of 3.
- Bronchial challenge (histamine/methacholine): FEV1 fall of **20% or more**. Others: raised FeNO, blood/sputum eosinophils.

Drugs

- **ICS = mainstay**; every controller must contain ICS; never LABA alone. Side effects: oral candidiasis, hoarseness, pneumonia at high dose.
- **ICS-formoterol** = preferred reliever and pre-exercise prophylaxis; Steps 1-2 = as-needed low-dose ICS-formoterol.
- OCS (prednisolone 40 mg 5-7 days) for exacerbations only. LTRA (montelukast) less effective than ICS; good for rhinitis; caution in neuropsychiatric illness.
- Beta-2 agonist side effects: tachycardia, tremor, anxiety, hypokalemia. Theophylline/aminophylline should **not** be used. IV MgSO4 only in severe exacerbations.

Control and severe asthma

- Cycle: confirm diagnosis, assess control, check triggers/comorbidities (GERD, rhinitis, obesity, smoking) and inhaler technique, adjust, review in 2-3 months; step down after 3 months of good control.
- Severe asthma with Th2 inflammation (sputum eos 2% or more, FeNO 20 ppb or more, blood eos 150/microL or more): biologics (omalizumab, benralizumab, dupilumab). Otherwise LAMA, azithromycin, bronchial thermoplasty.

Phenotypes and exacerbations

- Exercise-induced: symptoms 15-30 min after exercise. Aspirin-induced: 1-2 h after aspirin/NSAID, with rhinitis and sinusitis; avoid NSAIDs. Occupational: low PEF on workdays; avoid exposure.
- Severe attack: confusion, low BP, RR > 30, SpO2 < 90%, silent chest = ICU. Treat early with OCS, SABA + SAMA, oxygen to SpO2 > 93%.
- Discharge when SpO2 > 94% on air and PEF > 60-80%; follow up in 1-2 weeks.

Source: Asthma Lecture [005], Asthma LGW [005], Asthma CBD [005]

PART 1: LECTURE

COPD [009] (lecture and case)

Topic: COPD, asthma, wheezy chest

Definition and pathophysiology

- Heterogeneous condition: chronic respiratory symptoms from airway and/or alveolar abnormalities with **persistent, often progressive** airflow obstruction.
- Risk factors: smoking (about 50% of smokers), biomass fuel, alpha-1 antitrypsin deficiency, impaired lung development, childhood infection, long-standing asthma.
- Air trapping and V/Q mismatch cause low O2 and high CO2; chronic hypoxia leads to pulmonary hypertension and cor pulmonale.

Diagnosis = clinical + spirometry

- Starts in the 40s; persistent, progressive dyspnea is characteristic.
- **Post-bronchodilator FEV1/FVC < 70% is mandatory.** Signs have low sensitivity.
- Emphysema: thin, pursed lips, barrel chest, distant breath sounds, "pink". Chronic bronchitis: cough/sputum, obese, edema, cyanosis.
- CXR: hyperinflation, flat diaphragm, ribbon-shaped heart. CBC: secondary polycythemia.
- Screen all patients once for alpha-1 antitrypsin. Blood eos > 300 = ICS benefit.

Assessment and stable treatment

- Assess FEV1, mMRC dyspnea, exacerbations in the last year (2 or more, or 1 admission = high risk).
- All: smoking cessation, vaccines, rehabilitation.
- High exacerbation risk: LABA + LAMA; LAMA reduces exacerbations more than LABA.
- Exacerbations on LABA + LAMA with eos 300/microL or more: add ICS (triple). ICS + LABA alone not encouraged; avoid ICS if eos < 100, recurrent pneumonia or TB history.
- Still exacerbating on triple: azithromycin (250 mg/day for 1 year; arrhythmia, resistance) or roflumilast (needs **FEV1 < 50%**). Dupilumab blocks IL-4/IL-13.
- Theophylline not recommended unless inhalers unavailable: arrhythmia, convulsions, GI upset.

Oxygen

- LTOT 18-24 h/day if PaO2 < 55 mmHg (SpO2 < 88%), or < 60 mmHg with right heart failure or polycythemia; reassess every 2-3 months.
- High PCO2 (type II failure): target **SpO2 88-92%**; excess O2 can cause respiratory depression.

Acute exacerbation

- Worsening over < 14 days, usually infection. Exclude PE, pneumothorax, ACS.
- Antibiotics cover H. influenzae, S. pneumoniae, M. catarrhalis (ampicillin, macrolide, doxycycline); SABA + SAMA nebulizer; prednisolone 40 mg for 5 days; controlled O2; NIV/invasive ventilation if needed.
- Discharge when stable; check inhaler technique and O2 need; follow up in 1-4 weeks.

Source: COPD Lecture [009], COPD Case History [009]

PART 1: LECTURE

Wheezy chest [047] LGW

Topic: COPD, asthma, wheezy chest

Wheeze types

- Wheeze = continuous musical sound from vibration of narrowed airway walls; high or low pitched; inspiratory or expiratory.
- **All wheezes are obstruction, but not all wheezes are asthma.** Extra-thoracic obstruction limits inspiration; intra-thoracic limits expiration.
- **Stridor**: high-pitched; supraglottis to trachea (e.g. vocal cord dysfunction); often heard without stethoscope.
- **Rhonchi**: low-pitched, snoring-like; obstruction/secretions in large airways (COPD, bronchiectasis).

Causes

- Upper airway: post-nasal drip, vocal cord dysfunction, anaphylaxis (angioedema), foreign body, tracheal tumor.
- Lower airway: asthma, COPD, bronchiectasis, aspiration, heart failure ("cardiac asthma"), bronchogenic carcinoma (**localized wheeze**).
- Work-up: history, examination, spirometry with reversibility, challenge test, CXR/CT chest/CT neck, bronchoscopy.

Acute asthma

- Spirometry: usually too dyspneic. ABG only if SpO2 < 92%, life-threatening asthma or another diagnosis (e.g. DKA).
- CXR only if pneumothorax, pneumonia, life-threatening asthma or poor response.
- Treat: nebulized beta-2 agonist + ipratropium, prednisolone 20-40 mg for 5-7 days. **No routine antibiotics.**
- Discharge: symptoms improved, PEF > 60-80%, SpO2 > 94% on air, home support. Give reliever, start/step up controller, check technique, follow up in 2-7 days.

Acute exacerbation of COPD

- Older smoker, worsening cough and wheeze, prolonged expiration, hypoxemia.
- ABG example: pH 7.38, PaCO2 55, HCO3 32 = **compensated respiratory acidosis** (type II respiratory failure).
- CXR: hyper-expanded chest. Target SpO2 **88-92%** using a **Venturi mask**.
- Antibiotics: amoxicillin/clavulanate, macrolide or tetracycline. Sputum culture if frequent exacerbations or ventilated (resistant organisms, e.g. Pseudomonas).

Stridor plus wheeze

- Inspiratory stridor with expiratory wheeze in a smoker, with a paratracheal mass displacing the trachea = central airway compression; image with CT neck/chest and bronchoscopy.

Source: Wheezy Chest LGW [047]

PART 1: LECTURE

Interstitial lung disease [095] (lecture and LGW)

Topic: Interstitial lung disease

Basics

- Over 150 disorders with inflammation and fibrosis of the interstitium (between alveolar epithelium and capillary endothelium).
- Exposure-related: pneumoconioses (asbestosis, silicosis, coal), drugs (amiodarone, methotrexate, nitrofurantoin), connective tissue disease, radiation.
- IPF after 50; IPF and PLCH in smokers; sarcoidosis and HP less common in smokers. HP in bird breeders/farmers.

Clinical and tests

- Dyspnea is the main symptom; dry cough; clubbing; **Velcro inspiratory crackles**, basal first. Mid-inspiratory squeaks = HP.
- CXR can be misleadingly normal; **HRCT is the gold standard**.
- PFTs: **restrictive**, low TLC, RV, VC; normal or high FEV1/FVC; **low DLCO**. Resting hypoxemia from V/Q inequality; 6-minute walk unmasks desaturation.
- BAL clues: eos > 50% = eosinophilic pneumonia; rising RBCs + hemosiderin macrophages = DAH; milky fluid + foamy macrophages = PAP; Langerhans cells > 3%, CD1a/S100 = PLCH; CD4/CD8 < 1 = HP; 2.5 or more = sarcoidosis.
- Surgical biopsy preferred in most ILD; avoid honeycomb areas.

Patterns to recognize

- IPF: basal honeycombing, traction bronchiectasis; **pirfenidone**.
- Cysts: LAM (young women, thin-walled; **sirolimus, avoid pregnancy**); PLCH (smokers, bizarre shapes; **smoking cessation**, cladribine).
- PAP: **crazy paving**; whole-lung lavage, GM-CSF.
- DAH (e.g. GPA): alveolar filling; **plasmapheresis**.
- Radiation pneumonitis: ground glass on the irradiated side.

Sarcoidosis

- Non-caseating granulomas; age 20-40, female, black patients; lung in > 90%.
- **Löfgren**: fever, arthralgia, erythema nodosum, bilateral hilar nodes; no investigation, NSAIDs/rest only. **Heerfordt**: anterior uveitis, fever, parotid enlargement, facial palsy.
- CXR stages: 1 hilar nodes; 2 nodes + lung; 3 lung only; 4 fibrosis (not progressive).
- Hypercalcemia (10-20%) from macrophage vitamin D; ACE raised in > 75%, used for monitoring.
- Definitive diagnosis: biopsy with non-caseating granulomas. Prednisolone 0.5 mg/kg first line; add methotrexate, azathioprine or hydroxychloroquine.

General treatment

- Smoking cessation, oxygen, rehabilitation, vaccines, remove the offending agent; lung transplant for end-stage disease.

Source: ILD Lecture [095], ILD LGW [095]

PART 1: LECTURE

Occupational lung disease [109] LGW

Topic: Interstitial lung disease

Basics and general care

- Pneumoconiosis: inhaled dust/fibres taken up by alveolar macrophages → inflammation → lung fibrosis. Diagnosis needs a history of exposure.
- Primary prevention = reduce exposure. General measures: influenza + pneumococcal vaccines, stop smoking, bronchodilators, antibiotics for infections, oxygen for hypoxaemia.

Asbestos

- Exposure: mining/milling, boiler and pipe insulation, brake linings, roofing/floor products, shipyards (plumbers, pipe fitters).
- Pleural disease is most common: pleural plaques (localized calcified thickening at diaphragm level, usually incidental), diffuse pleural thickening, effusion, rounded atelectasis, mesothelioma.
- Rounded atelectasis: folded scarred pleura traps lung; comet tail sign; DD lung cancer (pseudotumour).
- Asbestosis: fibrosis of LOWER zones, 20-30 years after exposure (< 10 years if very heavy). Dyspnoea, cough, clubbing, bilateral inspiratory crackles, cor pulmonale.
- CXR reticulonodular opacities + honeycombing; restrictive PFTs (low volumes, low FVC, normal ratio), low DLCO, raised A-a gradient.
- Biopsy/sputum: ferruginous bodies (golden-brown barbell rods, Prussian blue).
- No specific treatment. Asbestos + smoking are synergistic for bronchogenic carcinoma (~75x risk): smoking cessation is key.

Silica

- Mining, quarrying, drilling, tunnelling, sandblasting, cement, glass and pottery. Pathology: silicotic nodule (concentric hyalinized collagen).
- Chronic (> 15 years): 1-10 mm UPPER lobe nodules → progressive massive fibrosis (PMF, > 10 mm); eggshell calcification of hilar nodes is characteristic.
- Acute (within months of massive exposure): lower-zone alveolar filling, rapid hypoxaemic failure, death.
- No effective therapy. TB complicates all forms → yearly tuberculin test.

Coal

- Coal macule (dust-laden macrophages round respiratory bronchioles) with focal emphysema; coal nodules are larger and palpable.
- Simple CWP: small rounded opacities < 1 cm, upper zones; PMF > 1 cm. Increased COPD risk. No cure, symptomatic only.
- Clubbing and crackles are NOT features: if present, look for another disease.
- Caplan syndrome: RA + coal/silica dust → nodular opacities, RF positive; treat as PMF + rheumatology protocol.

Source: Occupational Lung Disease LGW [109]

PART 1: LECTURE

Respiratory emergencies [105]

Topic: Respiratory emergencies

Red flags of life-threatening distress

- Altered mental state, severe cyanosis, pallor/sweating, 1-2 word dyspnoea, HR > 130, accessory muscles/retractions, absent breath sounds, stridor.

Pulmonary embolism: diagnosis

- Strong risk factors: hip/knee replacement, lower limb fracture, major trauma, spinal cord injury, previous VTE, recent admission (HF, AF, MI); also thrombophilia, OCP.
- Classic: sudden dyspnoea, pleuritic pain, tachycardia, hypoxia, often normal chest exam. ABG: low PaO2 (V/Q mismatch), low PaCO2.
- CXR usually normal; Hampton's hump (wedge opacity), Westermark, Fleischner, Palla's sign.
- D-dimer: good negative test (normal = excludes PE); raised value non-specific (cancer, hospital, infection, inflammation, pregnancy).
- CTPA = method of choice. V/Q scan (low radiation, no contrast): pregnancy, young women, contrast anaphylaxis, severe renal failure, low-probability outpatient with normal CXR.
- Leg compression ultrasound: incompressible vein = clot. Simplified PESI 0 = 1% vs ≥ 1 = 10.9% 30-day mortality.

Pulmonary embolism: treatment

- O2 to SaO2 93-98%; RV failure: fluid ≤ 500 ml, norepinephrine, dobutamine, ECMO.
- High-risk (shock): thrombolysis (rtPA, streptokinase, urokinase); alternatives catheter-directed treatment, surgical embolectomy. After thrombolysis in arrest, continue CPR 60-90 min.
- Absolute CI: stroke, CNS tumour, trauma/surgery/head injury < 3 weeks, bleeding diathesis, active bleeding. Relative: peptic ulcer, oral anticoagulant, pregnancy.
- Start LMWH/fondaparinux/UFH while awaiting tests if probability high/intermediate. Warfarin overlap ≥ 5 days and INR 2-3 for 2 days. NOACs: dabigatran (thrombin), apixaban/edoxaban/rivaroxaban (Xa).
- IVC filter: VTE + absolute CI to anticoagulation; recurrent PE despite anticoagulation; high-risk prophylaxis.

ARDS (Berlin)

- Within 1 week of insult (e.g. peritonitis, haemorrhage, burns); bilateral opacities not explained by effusion/collapse/nodules; not cardiac/fluid (echo, PAWP < 18); PaO2/FiO2 ≤ 300.
- Treat cause, neutral fluid balance, high FiO2, inhaled NO/epoprostenol, neuromuscular blockade, early high-dose steroids.

Atelectasis

- Types: obstructive (mucus plug, tumour, FB, tube), compressive (effusion, pneumothorax), cicatricial (scarring, post-TB), adhesive (no surfactant, ARDS).
- Collapse signs: retracted side, trachea TOWARDS lesion, dull, absent breath sounds. Postoperative: poor cough from pain, aspiration.
- Luftsichel sign (air crescent) = LUL collapse. DD of a dull silent side: effusion, fibrosis, collapse.

Source: Respiratory Emergencies [105]

PART 1: LECTURE

Respiratory cardinal symptoms [108]

Topic: Respiratory symptoms, presentations and clinical reasoning

Cough (six cardinal symptoms: cough, sputum, dyspnoea, chest pain, haemoptysis, wheeze)

- Acute < 3 weeks (bronchitis, pneumonia, PE); subacute 3-8 weeks (post-infection, asthma); chronic > 8 weeks (ILD, COPD).
- Larynx: painful, hoarse, stridor. Trachea: metallic, retrosternal soreness. Pneumonia: dry then productive. TB: weeks-months, blood-tinged. ILD: dry, persistent. Pleura: dry with sharp pain.
- GERD: VERY common; after meals, worse supine, heartburn/bitter taste. LVF: supine with orthopnoea.
- ACE inhibitors: dry cough in ~10%, 3-6 months after starting, more in women (bradykinin not broken down).
- Psychogenic: young female, barking, in front of audience; exclude organic causes. Complications: syncope, rib fracture, pneumothorax, hernia, incontinence.

Sputum

- Green = infection (myeloperoxidase); rusty = pneumococcus; red currant jelly = Klebsiella; black = coal miners, heavy smokers.
- 100 ml/day = bronchorrhoea: lung abscess, bronchiectasis, alveolar cell carcinoma. Offensive = anaerobes. Mucolytics: acetylcysteine, bromhexine.

Dyspnoea

- Sudden: PE, pneumothorax, foreign body, MI. Hours-days: pneumonia, asthma/COPD exacerbation. Weeks-months: ILD, heart failure. Grade with mMRC 0-4.
- Orthopnoea: LVF (commonest), ascites. PND ~2 h after sleep = cardiac; very early morning = asthma; worse on waking, eased by sputum = COPD.
- Better at weekends/holiday = occupational asthma; worse 15-30 min after exercise = exercise-induced asthma; tingling + light-headedness = psychogenic.

Chest pain

- Visceral pleura insensitive; pain from parietal pleura, chest wall, airways, diaphragm, mediastinum.
- Pleural: sharp, localized, worse on breathing/cough, refers to shoulder tip. Angina: crushing retrosternal, exertional, eased by rest/nitrate. Oesophageal: burning, meals, supine.
- Emergencies: MI, aortic dissection, PE, tension pneumothorax, mediastinitis, perforated viscus.

Haemoptysis and wheeze

- From below the vocal cords. False: GI blood (food particles, melaena) or nose/mouth/larynx.
- Common causes: cancer, TB, PE, bronchiectasis; also mitral stenosis, LVF, warfarin. Intermittent over years = bronchiectasis/COPD; daily = cancer, TB, abscess; single + pain + dyspnoea = PE/infarction.
- Massive > 150 ml/attack (slide figure): suffocation risk; resuscitate, cough suppressant, bronchoscopy, bronchial artery embolization.
- Wheeze: high-pitched, expiratory, small airways. Rhonchi: low-pitched, both phases, change with cough.

Source: Respiratory Cardinal Symptoms [108]

PART 1: LECTURE

Chest history taking and examination [100]

Topic: Respiratory symptoms, presentations and clinical reasoning

History

- Complaint in patient's words (open question); present history: onset, course, duration, severity, aggravating/relieving, associated symptoms.
- Cough: diurnal/seasonal variation. Sputum: colour, volume (1 cup = 50 ml), odour. Dyspnoea: MRC, orthopnoea/PND. Wheeze: relation to work. Haemoptysis: volume, attacks. Chest pain: SOCRATES.
- Also constitutional (fever, sweats, weight/appetite loss), congestive (oedema, RUQ pain, ascites, jaundice) and pressure symptoms (dysphagia, hoarseness).
- Drugs: oestrogen (PE), beta-blockers (bronchospasm), ACE inhibitors (cough), methotrexate/amiodarone (fibrosis); home oxygen.
- Social: smoking, cannabis, birds/pets, full occupational history. Family: atopy, cancer, TB. Travel: TB areas, long-haul flights (PE). Finish with ICE.

General examination

- Position 45 degrees, exposed to waist, examiner on right. RR normal 14-20; temp 36.5-37.2 C.
- Distress: accessory muscles, intercostal recession, pursed lips (stop early airway closure in expiration), nasal flaring, tripod position.
- Central cyanosis: lips and tongue. Hands: 3 Cs (clubbing, cyanosis, CRT) and 3 Ts (tremor, tar staining, temperature).
- Clubbing: 1st nail angle lost, 2nd parrot beak, 3rd drumstick, 4th hypertrophic osteoarthropathy. Flapping tremor = CO_2 retention.
- Legs: oedema (cor pulmonale), calf tenderness (DVT).

Inspection and palpation

- Normal AP:transverse = 5:7; barrel chest in emphysema. Limited side = abnormal side.
- Trachea TOWARDS upper lobe collapse/fibrosis or lobectomy; AWAY from tension pneumothorax or massive effusion.

Percussion

- Hyperresonant = pneumothorax; impaired = fibrosis; dull = consolidation, mass, collapse; stony dull = effusion.
- Kronig's isthmus dull: apical TB fibrosis, collapse, Pancoast tumour, consolidation, abscess, apical effusion.
- Traube's area dull: splenomegaly, big left liver lobe, tense ascites, left effusion, full stomach. Large: splenectomy, pneumothorax, pneumoperitoneum.

Auscultation

- Normal: vesicular (inspiration > expiration, no gap), equal, no added sounds.
- Bronchial breathing (3 Cs): consolidation, collapse with patent bronchus, superficial cavity; with increased VR (bronchophony, whispering pectoriloquy).
- Decreased VR and air entry: effusion, pneumothorax, collapse, mass; diffuse: emphysema, severe asthma, obesity.
- Wheeze: musical, louder on expiration; inspiratory = severe narrowing. Coarse early-mid crackles: oedema, bronchiectasis, abscess. Fine late: ILD, LVF, pneumonia.

Source: History and examination lecture [100]; student notes; history and examination checklists

PART 1: LECTURE

Pulmonary function and spirometry [062]

Topic: Respiratory symptoms, presentations and clinical reasoning

Spirometry basics

- Volume-time curve shows FVC and FEV1; flow-volume loop shows FVC, FEV1, FER (FEV1/FVC x 100) and PEF. Normally about 80% of FVC is blown out in the first second.
- Uses: dyspnoea work-up, pre-operative assessment, monitoring disease, occupational exposure, disability, response to treatment.
- Spirometry cannot measure RV, so it cannot give TLC or FRC. Volumes are in litres at BTPS.
- Results are compared with predicted values from age, sex, height, weight (and ethnicity).

Doing the test

- Before: no bronchodilators, no smoking for 1 h, no heavy exercise for 30 min, no heavy meal for 2 h, loose clothing.
- Nose clip, tight seal; full inspiration, blow hard and fast to RV. Effort-dependent: coach the patient.
- At least 3 acceptable blows (no more than 8); record best FVC, FEV1, PEF.
- Acceptable: quick forceful start, no cough or leak, smooth curve, exhalation at least 6 s with no volume change in the last 1 s; efforts agree within 5% or 100 mL.

Patterns

- Obstructive: low FEV1/FVC, low FEV1 and FEF25-75, normal or low FVC, normal or high TLC; scooped expiratory curve (asthma). Early collapse of flow = advanced obstruction (COPD).
- Restrictive: normal or high FEV1/FVC, low FVC, low TLC (confirm TLC in PFT lab); tall narrow loop (pulmonary fibrosis).
- Variable extrathoracic upper airway obstruction (goitre): inspiratory loop most flattened.
- Positive bronchodilator response: FEV1 up by 12% and 200 mL (asthma).
- Raised RV = air trapping; hyperinflation = TLC above 120%.
- Normal ranges: FEV1, FVC, TLC 80-120%; RV 75-120%; DLCO above 60%.
- COPD (FEV1/FVC below 0.7) by FEV1: GOLD 1 80% or more, 2 50-79%, 3 30-49%, 4 below 30%.

Lung volumes and gas transfer

- Helium dilution: FRC of ventilated lung only. Body plethysmography: thoracic gas volume including non-ventilated areas.
- DLCO uses CO (single breath-hold); helium gives alveolar volume (VA); KCO = TLCO / VA.
- DLCO low: ILD, emphysema, pneumonectomy, anaemia, V/Q mismatch. DLCO high: pulmonary haemorrhage (Goodpasture), polycythaemia.
- Restriction on spirometry: order DLCO next. Bird exposure + restriction + low DLCO = hypersensitivity pneumonitis: remove antigen + oral steroids.

Source: Pulmonary Function [062] slides and handout

PART 1: LECTURE

Pleural diseases [117] LGW

Topic: Respiratory symptoms, presentations and clinical reasoning

Pleural effusion: clinical

- Normal pleural space has 1-15 mL fluid. Dyspnoea is the commonest symptom; pleuritic pain suggests PE or inflammation; constant dull ache suggests chest wall invasion (carcinoma, mesothelioma).
- Signs from about 300 mL: stony dullness, reduced breath sounds and vocal resonance. Above 1000 mL: mediastinal shift away.
- Shift TOWARDS the effusion = bronchial obstruction by an endobronchial lesion (usually malignancy).

Causes and fluid analysis

- Transudates: heart failure, constrictive pericarditis, cirrhosis, hypoalbuminaemia, nephrotic syndrome, SVC syndrome.
- Exudates: parapneumonic, TB, malignancy, RA/SLE, PE, asbestos, pancreatitis, trauma, oesophageal perforation, radiation, drugs.
- Light's criteria (any one = exudate): protein ratio above 0.5; LDH ratio above 0.6; pleural LDH above 2/3 upper normal serum LDH.
- Protein above 2.9 g/dL = exudate. LDH above 1000 = empyema or malignancy. pH below 7.2 and glucose below 30 = empyema. Glucose 30-50 = malignancy or TB.
- Milky + TG above 110 mg/dL = chylothorax. Pleural Hct above 50% of blood Hct = haemothorax.
- Lymphocytes above 85% = TB, lymphoma. Eosinophils above 10% = air/blood in pleura, PE with infarction, parasites, drugs. WBC above 1000/mm3 = exudate.
- Malignant effusion: haemorrhagic, massive, rapidly reaccumulating.

Imaging and thoracentesis

- PA CXR detects 200-300 mL (blunted angle); lateral decubitus 75-100 mL. D-shaped opacity suggests empyema.
- Tap 2 rib spaces below dullness, guided by US or CT; check position with small needle first; give oxygen.
- Remove at most 1-1.5 L at once (re-expansion oedema, pneumothorax). Post-tap expiratory CXR to exclude pneumothorax.
- Relative contraindications: very small effusion, bleeding disorder or anticoagulation, mechanical ventilation, skin disease at site.
- Empyema: large-bore (20 F) chest tube is a must.

Haemothorax and pneumothorax

- Haemothorax: trauma or procedures. Large-bore drain, cross-match, correct clotting; above 1000 mL or above 200 mL/h for 3 h = call thoracic surgeon.
- Primary spontaneous: thin young men, no lung disease (Marfan, Ehlers-Danlos). Secondary: COPD, asthma, CF.
- Signs: reduced expansion, hyper-resonance, reduced breath sounds. CXR: no lung markings beyond a sharp lung edge.
- Tension: hypotension, tachycardia, raised JVP, shift away. Immediate needle decompression, 2nd intercostal space, midclavicular line, then chest tube.

Source: Pleural Diseases [117] LGW

PART 1: LECTURE

Chest case reviews [098] and cases W, X, Y, Z [021, 022]

Topic: Respiratory symptoms, presentations and clinical reasoning

Airways disease

- Child with recurrent wheeze, cough, dyspnoea, pulsus paradoxus = acute asthma. NSAIDs can trigger attacks. Clubbing is not a feature of asthma.
- COPD = airflow limitation that is not fully reversible (e.g. FEV1/FVC 47%, no reversibility, smoker). Moderate = FEV1 50-79% predicted.
- COPD does not cause clubbing: new clubbing in a smoker suggests bronchogenic carcinoma. Respiratory causes: lung cancer, mesothelioma, empyema, CF.
- Raised JVP, hepatomegaly and oedema in COPD suggest cor pulmonale.

Blood gases and respiratory failure

- Type 1 failure: PO2 below 60 mmHg with normal or low PCO2 (e.g. PE). Type 2: raised PCO2.
- pH 7.61, PCO2 22, HCO3 24 = uncompensated respiratory alkalosis (anxiety hyperventilation).
- pH 7.32, PCO2 60, HCO3 24 = respiratory acidosis (COPD exacerbation).

PE and ARDS

- After a long flight: Westermark sign, Hampton's hump, type 1 failure = PE.
- D-dimer is sensitive, not specific. Start LMWH + warfarin (INR 2-3) while CTPA is awaited.
- ARDS: acute, bilateral infiltrates, PaO2:FiO2 below 300, no heart failure. Mild 200-300, moderate 100-200, severe 100 or less. Aspiration is a trigger.

Infection and granulomas

- CURB-65: confusion, urea above 7, RR 30 or more, low BP, age 65 or more (one point each).
- Fibrocavitary upper lobe lesion, night sweats, haemoptysis, weight loss = TB. Treat 2 months RHZE then 4 months RH; check LFTs. Rifampicin turns urine orange-red.
- Caseating granulomas = TB; non-caseating granulomas + bilateral hilar nodes = sarcoidosis (restrictive PFTs, low DLCO).
- Asbestos (stripping ship insulation) + basal fine crackles = asbestosis.

Pleura and lung cancer

- High protein and LDH = exudate (e.g. bronchogenic carcinoma); heart failure, cirrhosis, nephrotic and SVC syndrome cause transudates.
- Empyema fluid: pH below 7.2, high LDH, low glucose.
- Suspected pneumothorax: chest X-ray first.
- Pancoast tumour: Horner syndrome (ptosis, miosis, anhidrosis); no exophthalmos.

Source: Chest Cases Review [098]; Cases WX and YZ [021, 022]

PART 2: TOPIC SUMMARY

Infections (pneumonia, TB, abscess, bronchiectasis)

Pneumonia

- CAP (community), HAP (48 h or more after admission), VAP (48 h or more on a ventilator) have different organisms and antibiotics. Commonest CAP organism: S. pneumoniae.
- Severity: CURB-65 (confusion, urea over 7, RR 30 or more, low BP, age 65 or more). 0-1 home; 2 ward; 3-5 or severe criteria = ICU.
- Outpatient, healthy: amoxicillin, doxycycline or macrolide. Inpatient: beta-lactam + macrolide. Add anti-MRSA or anti-Pseudomonas only with risk factors.
- Atypicals: Mycoplasma (young, headache, no cell wall, macrolide); Legionella (travel, water aerosols, confusion, diarrhoea, macrolide).

Tuberculosis

Step	Key facts
Diagnose	3 early-morning sputa; Ziehl-Neelsen or fluorescent smear; culture (slow, 4-8 weeks, gives resistance); PCR rpoB = rifampicin resistance
Latent TB	TST (read at 48-72 h; cut-offs 5, 10, 15 mm by risk) or IGRA (not affected by BCG)
Treat	2 months HRZE then 4 months HR; meningitis 9-12 months; bone 9 months
Side effects	Isoniazid neuropathy (pyridoxine), hepatitis (H, R, Z); ethambutol vision (red-green first); rifampicin orange urine; streptomycin ear damage (avoid in pregnancy)
Monitor	Sputum at 2 months and at the end; LFTs

- Extrapulmonary TB: miliary (2-3 mm nodules), meningitis (most serious; CSF lymphocytes, low glucose, high protein), Pott's spine (paraplegia), lymphadenitis. Steroids for TB pericarditis and meningitis.
- HIV: atypical TB; start ART 2-8 weeks after TB treatment if CD4 under 200.
- NTM: MAC commonest; child with one-sided neck node = cure by excision; fish tank granuloma = M. marinum. MAC lung: macrolide + ethambutol + rifampicin.

Suppurative lung disease

- Lung abscess: cavity with air-fluid level, usually aspiration (anaerobes). Antibiotics 4-6 weeks; drainage if failing; lobectomy if over 6 cm.
- Bronchiectasis: permanent bronchial dilatation, large purulent sputum, clubbing, coarse crackles. HRCT = gold standard (signet ring). First Pseudomonas: ciprofloxacin then inhaled colistin; frequent exacerbations: long-term azithromycin.

PART 2: TOPIC SUMMARY

Lung malignancies

Bronchogenic carcinoma at a glance

- Old smoker with chest symptoms = cancer until proven otherwise. Central = squamous (cough, haemoptysis); peripheral = adenocarcinoma (pleural pain).
- Local spread: Pancoast tumour (shoulder pain, hand wasting, Horner's: ptosis, miosis, anhidrosis); SVC obstruction (facial plethora; classically small cell); hoarseness (left recurrent laryngeal nerve); localised wheeze.

Cell type	Paraneoplastic clue
Small cell	SIADH (low sodium), ectopic ACTH, Lambert-Eaton
Squamous	Hypercalcaemia (PTH-like peptide)
Large cell / adeno	Gynaecomastia; clubbing and hypertrophic osteoarthropathy

- Tissue is the issue: bronchoscopy for central, CT-guided biopsy for peripheral; stage with CT chest, abdomen, brain and PET-CT.
- TNM: T1 up to 3 cm, T2 3-5 cm, T3 5-7 cm, T4 over 7 cm or mediastinal invasion; N1 hilar, N2 same-side mediastinal, N3 opposite side or supraclavicular; M1a opposite lung or malignant effusion.
- Non-small cell: surgery if resectable. Small cell: limited versus extensive, chemotherapy (surgery usually not possible).
- Clubbing in a smoker with COPD is not from COPD: look for cancer.

PART 2: TOPIC SUMMARY

COPD, asthma, wheezy chest

Asthma versus COPD

	Asthma	COPD
Key word	Variable	Persistent
Spirometry	FEV1/FVC under 80% with reversibility (FEV1 up 12% and 200 mL)	Post-bronchodilator FEV1/FVC under 70%
Mainstay	Inhaled steroid (never LABA alone); ICS-formoterol reliever	LABA + LAMA; add ICS if eosinophils 300 or more with exacerbations
Acute attack	SABA + ipratropium, prednisolone, oxygen to SpO2 over 93%	SABA + SAMA, prednisolone 40 mg 5 days, antibiotics, oxygen 88-92% via Venturi
Danger signs	Silent chest, confusion, SpO2 under 90%	Type 2 failure (high CO2)

- COPD long-term oxygen: PaO2 under 55 mmHg (or under 60 with cor pulmonale or polycythaemia), 18-24 h a day. Screen once for alpha-1 antitrypsin.
- Asthma phenotypes: exercise-induced, aspirin-induced (avoid NSAIDs), occupational (better at weekends).
- Not every wheeze is asthma: stridor = upper airway; localised wheeze = tumour or foreign body; cardiac asthma = heart failure.

PART 2: TOPIC SUMMARY

Interstitial lung disease

Shared features

- Dyspnoea, dry cough, clubbing, fine Velcro crackles at the bases. Restrictive PFTs with low DLCO. HRCT is the gold standard.

Disease	Clue	Treatment
IPF	Over 50, basal honeycombing	Pirfenidone
Sarcoidosis	Young, bilateral hilar nodes, non-caseating granulomas, raised ACE and calcium	Prednisolone; Löfgren needs only NSAIDs
Hypersensitivity pneumonitis	Birds or farming; mid-inspiratory squeaks; BAL CD4/CD8 under 1	Remove antigen, steroids
LAM	Young woman, thin-walled cysts	Sirolimus, avoid pregnancy
PLCH	Smoker, odd-shaped cysts	Stop smoking

Occupational lung disease

- Asbestos: pleural plaques (commonest), asbestosis in lower zones, mesothelioma; with smoking, lung cancer risk multiplies.
- Silica: upper lobe nodules, eggshell calcification of hilar nodes, TB risk (yearly test).
- Coal: simple CWP under 1 cm, PMF over 1 cm; clubbing and crackles are not features. Caplan = RA + dust nodules.

PART 2: TOPIC SUMMARY

Respiratory emergencies

Pulmonary embolism

- Sudden dyspnoea, pleuritic pain, hypoxia, often clear chest. Low PaO2 with low PaCO2.
- D-dimer excludes when negative; CTPA confirms (V/Q in pregnancy, contrast allergy, renal failure).
- High risk (shock) = thrombolysis unless contraindicated. Otherwise anticoagulate (LMWH, DOAC, or warfarin overlap to INR 2-3).

Other emergencies

- Tension pneumothorax: hypotension, trachea away, hyper-resonant: immediate needle decompression (2nd intercostal space, midclavicular line), then chest tube.
- ARDS: within a week of an insult, bilateral opacities, not cardiac, PaO2/FiO2 300 or less (mild 200-300, moderate 100-200, severe 100 or less).
- Collapse: trachea pulled towards the lesion; Luftsichel sign = left upper lobe collapse.
- Massive haemoptysis: protect the good lung, bronchoscopy, bronchial artery embolisation.

PART 2: TOPIC SUMMARY

Respiratory symptoms, presentations and clinical reasoning

Symptoms

- Cough: acute under 3 weeks, chronic over 8 weeks; common hidden causes = reflux, ACE inhibitors (bradykinin), post-nasal drip.
- Sputum: rusty = pneumococcus; red currant jelly = Klebsiella; over 100 mL a day = bronchiectasis or abscess.
- Haemoptysis: cancer, TB, PE, bronchiectasis; first exclude blood from nose or stomach.

Examination findings

Finding	Trachea	Percussion	Breath sounds
Consolidation	Central	Dull	Bronchial, increased resonance
Effusion	Away (if large)	Stony dull	Reduced
Collapse	Towards	Dull	Reduced
Pneumothorax	Away (tension)	Hyper-resonant	Reduced
Fibrosis	Towards (upper lobe)	Impaired	Fine crackles

Pleural fluid

- Light's criteria: exudate if protein ratio over 0.5, LDH ratio over 0.6, or LDH over two-thirds of the upper normal. Transudates: heart, liver, kidney failure.
- Empyema: pH under 7.2, low glucose, high LDH: large-bore chest drain. Lymphocytes over 85% = TB or lymphoma. Remove at most 1-1.5 L at a time.

Spirometry

- Obstructive: low FEV1/FVC; restrictive: normal or high ratio with low FVC and TLC. Low DLCO = ILD or emphysema; high DLCO = pulmonary haemorrhage.
- ABG: type 1 failure = low O2, normal or low CO2; type 2 = high CO2.